Multiview latent world model

Current views





Selected views

Spatial encoder

Frozen

Future-feature predictor

Predicted features

View 1 Feature decoder

View 2 Feature decoder

Other views Feature decoder

Current multiview features

Shared latent action

Common inputs to every view decoder

Inverse model · training only

Current + true future features

Inverse dynamics encoder

Teacher latent action

Spatial compressor

Compression queries

Query-based aggregation

Camera embeddings + valid-view mask

Future context

Action expert

Robot state & Noisy actions

First 9 transformer layers

Action-token self-attention

Context cross-attention

Hidden state

Correction adapter

Residual addition

Correction code

Last 9 transformer layers

Action-token self-attention

Context cross-attention

Progress head

Linear + sigmoid

Per-step progress

Velocity field

10-step flow integration

50-step action chunk

Continuous action generation

Observation history

Earlier → Current



5 past + current

Selected frames shown

Motion-aware history encoder

Visual tokens

Predicted latent action

Policy inference

Qwen3-VL

2B parameters

Current context

Subtask instruction

Motion-aware history encoder



5 past + current; selected shown

Temporal kernel 2

First 4 shared blocks

3 paired representations

4 × 4 pool + learned queries

Pair summaries

Adjacent differences

Projection + position + layer normalization

Motion keys and values

Visual features

Gated cross-attention

+ residual fusion

Remaining blocks

+ visual merger

Visual tokens